Dr. Md Nadim

Curriculum Vitae

Faculty Member
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Career Objective

To secure a research-intensive postdoctoral position where I can extend my work in quantum and classical machine learning and software engineering, with a focus on AI-driven practical applications. I aim to develop reliable and high-quality intelligent systems, collaborate with academic and industry partners, and translate advanced learning methods into real-world solutions that improve the lives of people who need them most.

Research Interests

- **Empirical Software Engineering & Reliable AI Systems:** Study software quality, security, and scalability in both human-written and AI-generated code, with a focus on building trustworthy systems for real-world domains such as healthcare and safety-critical applications.
- **AI for Health & Predictive Analytics:** Apply machine learning and deep learning methods to health and mHealth data to support early prediction, personalized care, and better decision making in clinical settings.
- **Quantum-Enhanced Software Analytics:** Design hybrid quantum and classical machine learning models to handle complex software engineering tasks, with the goal of improving performance and enabling new solutions for large-scale and data-intensive problems.

Education

2020–2025: **PhD, Computer Science**, *Software Research Lab*, University of Saskatchewan, Canada.
Software bug prediction using Quantum and Classical machine learning algorithms, their execution, comparison, and challenges. Utilised IBM Quantum services, D-Wave Quantum Annealing systems, and large language models for experimenting with QML algorithms in real quantum machines and simulators.

2018–2020: **MSc, Computer Science**, *Software Research Lab*, University of Saskatchewan, Canada.

2006–2010 : **BSc, Computer Science & Engineering**, *Hajee Mohammad Danesh Science & Technology University*, Dinajpur, Bangladesh.

Publications

Journal Articles

2025 **Md Nadim**, Mohammad Hassan, Ashis Kumar Mandal, Chanchal K. Roy, Banani Roy, and Kevin A. Schneider. Comparative analysis of **quantum and classical support vector classifiers** for software bug prediction: an exploratory study. *Quantum Machine Intelligence*, volume 7, page 32, Mar 2025.

2025 Ashis Kumar Mandal, **Md Nadim**, Chanchal K. Roy, Banani Roy, and Kevin A. Schneider. **Quantum software engineering and potential of quantum computing** in software engineering research: a review. *Automated Software Engineering*, volume 32, page 27, Mar 2025.

2023 **Md Nadim** and Banani Roy. Utilizing source code syntax patterns to detect bug inducing commits using machine learning models. *Software Quality Journal*, volume 31, pages 775–807, Sep 2023.

2022 **Md Nadim**, Manishankar Mondal, Chanchal K. Roy, and Kevin A. Schneider. Evaluating the performance of clone detection tools in detecting cloned co-change candidates. *Journal of Systems and Software*, volume 187, page 111229, 2022.

2022 **Md Nadim**, Debajyoti Mondal, and Chanchal K. Roy. Leveraging structural properties of source code graphs for just-in-time bug prediction. *Automated Software Engineering*, volume 29, page 27, Mar 2022.

Conference Proceedings

2025 **Md Nadim**, Mohammad Hassan, Ashis Kumar Mandal, and Chanchal K. Roy. **Quantum Vs. Classical Machine Learning Algorithms** for Software Defect Prediction: Challenges and Opportunities . In *IEEE/ACM International Workshop on Quantum Software Engineering (Q-SE)*, pages 35–42, Los Alamitos, CA, USA, May 2025. IEEE Computer Society.

2024 Ashis Kumar Mandal, **Md Nadim**, Chanchal K. Roy, Banani Roy, and Kevin A. Schneider. Evaluating the performance of a **D-Wave Quantum Annealing System** for feature subset selection in software defect prediction. In *IEEE International Conference on Quantum Computing and Engineering (QCE)*, volume 02, pages 103–108, 2024.

2020 **Md Nadim**, Manishankar Mondal, and Chanchal K. Roy. Evaluating performance of clone detection tools in detecting cloned cochange candidates. In *IEEE 14th International Workshop on Software Clones (IWSC)*, pages 15–21, 2020.

2016 Shahnaj Parvin Shathi, Md. Delowar Hossain, **Md Nadim**, Sayed Golam Rasul Riayadh, and Tangina Sultana. Enhancing performance of naïve bayes in text classification by introducing an extra weight using less number of training examples. In *2016 International Workshop on Computational Intelligence (IWCI)*, pages 142–147, 2016.

2010 Ashis Kumar Mandal, Md. Delowar Hossain, and **Md Nadim**. Developing an efficient search suggestion generator, ignoring spelling error for high speed data retrieval using double metaphone algorithm. In *13th International Conference on Computer and Information Technology (ICCIT)*, pages 317–320, 2010.

Research Experience

University of Saskatchewan, Canada

Sep, 2018 – Apr, 2025 ***Graduate Researcher***.
I conduct research on software defect prediction and code clone detection, with a focus on using machine learning, deep learning, and quantum-inspired methods. My work explores source code structure, syntax patterns, and feature learning to improve bug prediction and software quality.

Advisor : **Dr. Chanchal K. Roy**, *Professor, Department of Computer Science*, University of Saskatchewan, Canada (*Personal Web-page*)

Teaching Experience

May 2025 – Present **Associate Professor, Department of CSE**, *Hajee Mohammad Danesh Science and Technology University (HSTU)*, Dinajpur, Bangladesh.
Courses Taught: Software Engineering (CSE-305); Fundamentals of Computer & Programming (CSE-101/102/107/108/132); Mathematical Analysis for Computer Science (CSE-361); Advanced Machine Learning (CSE-547)

2018 – 2025 **Study Leave, M.Sc. and Ph.D.**, *University of Saskatchewan*, Saskatoon, Canada.
September 1, 2018 to May 14, 2025

Feb 2012 – Aug 2018 **Lecturer/ Assistant Professor, Department of CSE**, *HSTU*, Dinajpur, Bangladesh.
I joined as a Lecturer on February 1, 2012, and was promoted to Assistant Professor on February 1, 2015. During this time, I continued teaching key undergraduate courses at HSTU, including Structured Programming (C), Object-Oriented Programming (C++ and Java), Data Structures, Algorithms, Operating Systems, Database Management Systems, and Pattern Recognition.

Teaching Assistantship @USASK, SK, Canada

Winter, 2024 **CMPT 340: Programming Language Paradigms**, *132 Hours.*

Fall, 2023 **CMPT 214: Programming Principles and Practice**, *88 Hours.*

Spring, 2023 **CMPT 145: Principles of Computer Science**, *66 Hours*.
Winter, 2023 **CMPT 280: Intermediate Data Structures and Algorithms**, *88 Hours*.
Winter, 2022 **CMPT 141: Introduction to Computer Science**, *88 Hours*.
Winter, 2021 **CMPT 470: Advanced Software Engineering**, *88 Hours*.
Fall, 2020 **CMPT 214: Programming Principles and Practice**, *88 Hours*.

Fellowships & Awards

2018 Dean's Scholarship, University of Saskatchewan
2010 Prime Minister Gold Medal, Bangladesh

Leadership and Community Service

Jan 21, 2024 Crowd Support (Watch Party), TEDx USASK, Canada
2020–2021 Vice President, CSGC, USASK, Canada
2019–2020 GSA Representative, CSGC, USASK, Canada
May 2019 Volunteer Lead (Data Science Workshop), DIGITIZED, USASK, Canada

Key Technical Skills

Programming Python, C, C++, Java
Tools Machine Learning, Deep Learning, Quantum Computing, High-Performance Computing, NumPy, Pandas, Scikit-learn, TensorFlow/PyTorch, Git
Quantum Algorithms Phase Estimation, Amplitude Amplification, Grover's Search, Shor's Algorithm, Hamiltonian Simulation, Qubitization, Quantum Error Correction (QEC)
Web & DB HTML5, JavaScript, PHP, SQL, MySQL
Scripting Linux, Shell Scripting

Certification Courses

2024 **Graduate Professional Skills Certificate @University of Saskatchewan**, *SK*, Canada.
The Graduate Professional Skills Certificate is a comprehensive, non-credit program for graduate students and postdoctoral researchers at the University of Saskatchewan, SK, Canada. Within this Certificate program, GPS 974 focuses on strength-based professional skills development, building reflective practice, and developing a professional portfolio for improving skills and growth.

2024 **Thinking Critically@University of Saskatchewan**, *SK*, Canada.
The goal of this class is to provide a supportive and challenging setting to develop the creative and critical thinking skills required for professional practice. GPS 984 focuses on foundational frameworks of thinking (often invisible to us) that are used for almost everything we do in our personal and professional lives.

Referees

Dr. Chanchal K. Roy
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